<IGBT Modules>

CM150MXUD-24T/CM150MXUDP-24T

HIGH POWER SWITCHING USE
INSULATED TYPE

MXUD



Collector current I_C ..... **150** A
Collector-emitter voltage V_{CES} ..... **1200** V
Maximum junction temperature T_{vjmax} ..... **175** °C

- Flat base type
- Copper base plate (Nickel-plating)
- RoHS Directive compliant
- Tin-plating pin terminals

MXUDP



Collector current I_C ..... **150** A
Collector-emitter voltage V_{CES} ..... **1200** V
Maximum junction temperature T_{vjmax} ..... **175** °C

- Flat base type
- Copper base plate (Nickel-plating)
- RoHS Directive compliant
- Tin-plating pressfit terminals

CIB (Converter+Inverter+Chopper Brake)

- UL Recognized under UL1557, File No. E323585

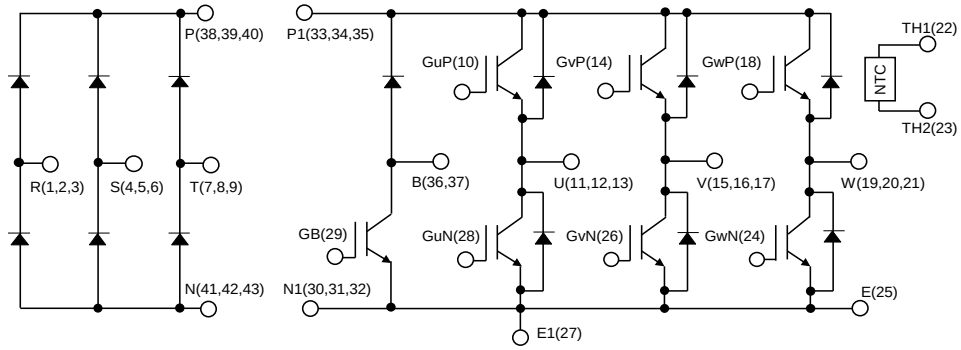
APPLICATION

AC Motor Control, Motion/Servo Control, Power supply, etc.

OPTION (Below options are available.)

- PC-TIM (Phase Change Thermal Interface Material) pre-apply (Note10)

INTERNAL CONNECTION



Terminal code

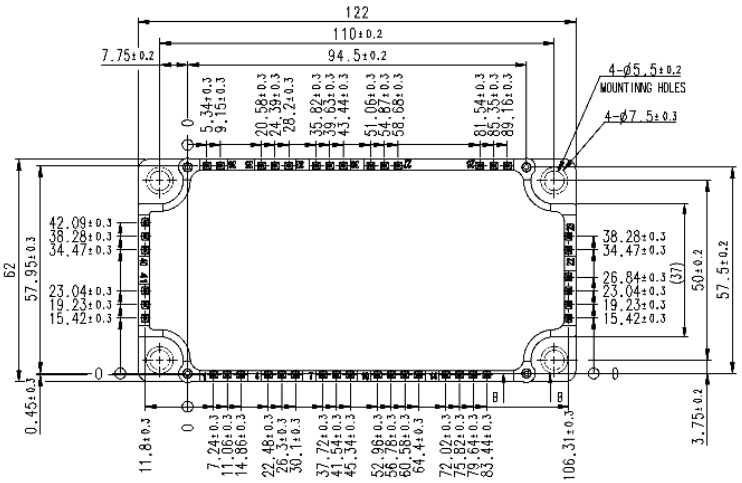
1	R	16	V	31	N1
2	R	17	V	32	N1
3	R	18	GwP	33	P1
4	S	19	W	34	P1
5	S	20	W	35	P1
6	S	21	W	36	B
7	T	22	TH1	37	B
8	T	23	TH2	38	P
9	T	24	GwN	39	P
10	GuP	25	E	40	P
11	U	26	GvN	41	N
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13	U	28	GuN	43	N
14	GvP	29	GB		
15	V	30	N1		

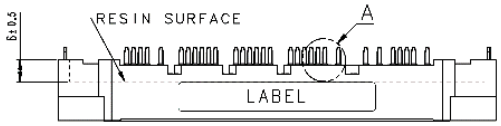
CM150MXUD-24T/CM150MXUDP-24T

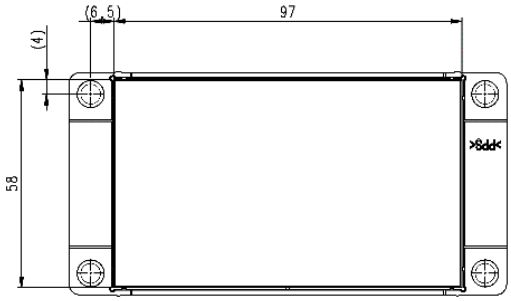
HIGH POWER SWITCHING USE
INSULATED TYPE

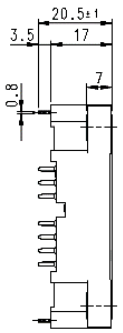
OUTLINE DRAWING **Dimension in mm**

MXUD

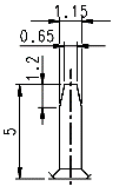




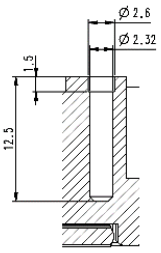




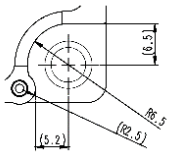
SECTION A



SECTION B



MOUNTING HOLES



Tolerance otherwise specified

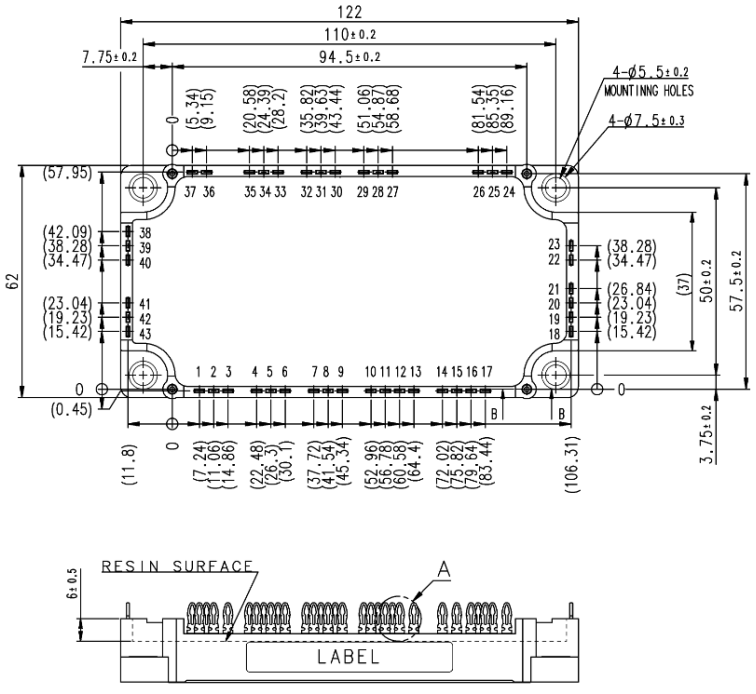
Division of Dimension		Tolerance
0.5	to 3	±0.2
over 3	to 6	±0.3
over 6	to 30	±0.5
over 30	to 120	±0.8
over 120	to 400	±1.2

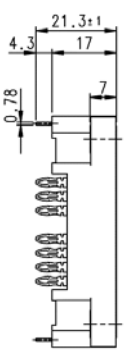
CM150MXUD-24T/CM150MXUDP-24T

HIGH POWER SWITCHING USE
INSULATED TYPE

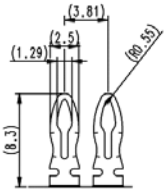
OUTLINE DRAWING **Dimension in mm**

MXUDP

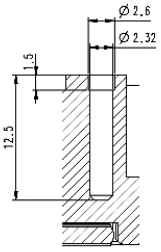


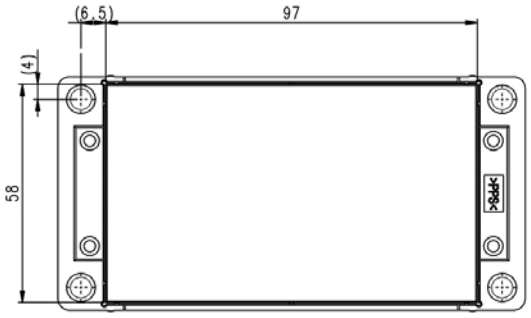


TERMINAL

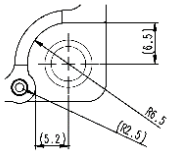


SECTION A

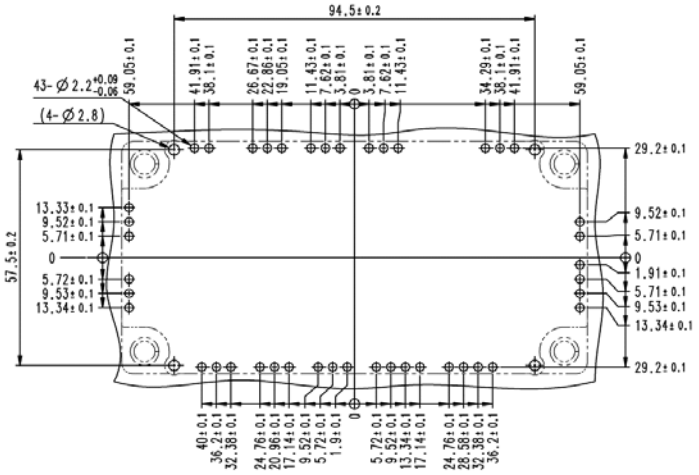




MOUNTING HOLES



PCB DRILL HOLE PATTERN



Tolerance otherwise specified

Division of Dimension		Tolerance
0.5	to 3	±0.2
over 3	to 6	±0.3
over 6	to 30	±0.5
over 30	to 120	±0.8
over 120	to 400	±1.2

MAXIMUM RATINGS (T_{vj}=25 °C, unless otherwise specified)

INVERTER PART IGBT/FWD

Symbol	Item	Conditions	Rating	Unit
V_{CES}	Collector-emitter voltage	G-E short-circuited	1200	V
V_{GES}	Gate-emitter voltage	C-E short-circuited	± 20	V
I_C	Collector current	DC, T_C=112 °C (Note2, 4)	150	A
I_{CRM}		Pulse, Repetitive (Note3)	300	
P_{tot}	Total power dissipation	T_C=25 °C (Note2, 4)	760	W
I_E (Note1)	Emitter current	DC (Note2)	150	A
I_{ERM} (Note1)		Pulse, Repetitive (Note3)	300	
T_{vjmax}	Maximum junction temperature	Instantaneous event (overload) (Note10)	175	°C

BRAKE PART IGBT/DIODE

Symbol	Item	Conditions	Rating	Unit
V_{CES}	Collector-emitter voltage	G-E short-circuited	1200	V
V_{GES}	Gate-emitter voltage	C-E short-circuited	± 20	V
I_C	Collector current	DC, T_C=119 °C (Note2, 4)	100	A
I_{CRM}		Pulse, Repetitive (Note3)	200	
P_{tot}	Total power dissipation	T_C=25 °C (Note2, 4)	565	W
V_{RRM}	Repetitive peak reverse voltage	G-E short-circuited	1200	V
I_F	Forward current	DC (Note2)	75	A
I_{FRM}		Pulse, Repetitive (Note3)	150	
T_{vjmax}	Maximum junction temperature	Instantaneous event (overload) (Note10)	175	°C

CONVERTER PART DIODE

Symbol	Item	Conditions		Rating	Unit
V_{RRM}	Repetitive peak reverse voltage	-		1600	V
E_a	Recommended AC input voltage	RMS		440	V
I_o	DC output current	3-phase full wave rectifying, T_C=125 °C (Note4)		150	A
I_{FSM}	Surge forward current	The sine half wave 1 cycle peak value,	T_{vj}=25 °C	1920	A
		f=60 Hz, non-repetitive	T_{vj}=150 °C	1536	
I^2t	Current square time	Value for one cycle of surge current	T_{vj}=25 °C	15360	A^2s
			T_{vj}=150 °C	9830	
T_{vjmax}	Maximum junction temperature	Instantaneous event (overload) (Note10)		150	°C

MODULE

Symbol	Item	Conditions	Rating	Unit
V_{isol}	Isolation voltage	Terminals to base plate, RMS, f=60 Hz, AC 1 min	2500	V
T_{Cmax}	Maximum case temperature	(Note4, 10)	125	°C
T_{vjop}	Operating junction temperature	Continuous operation (under switching) (Note10)	-40 ~ +150	°C
T_{stg}	Storage temperature	-	-40 ~ +125	

ELECTRICAL CHARACTERISTICS (T_{vj}=25 °C, unless otherwise specified)

INVERTER PART IGBT/FWD

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
I_{CES}	Collector-emitter cut-off current	$V_{CE}=V_{CES}$, G-E short-circuited		-	-	1.0	mA
I_{GES}	Gate-emitter leakage current	$V_{GE}=V_{GES}$, C-E short-circuited		-	-	0.5	μA
$V_{GE(th)}$	Gate-emitter threshold voltage	I_C=15 mA, V_{CE}=10 V		5.4	6.0	6.6	V
V_{CEsat} (Terminal)	Collector-emitter saturation voltage	I_C=150 A, V_{GE}=15 V, Refer to the figure of test circuit (Note5)	T_{vj}=25 °C	-	1.75	2.20	V
			T_{vj}=125 °C	-	2.10	-	
			T_{vj}=150 °C	-	2.20	-	
V_{CEsat} (Chip)		I_C=150 A, V_{GE}=15 V, (Note5)	T_{vj}=25 °C	-	1.50	1.75	V
			T_{vj}=125 °C	-	1.70	-	
			T_{vj}=150 °C	-	1.75	-	
C_{ies}	Input capacitance	V_{CE}=10 V, G-E short-circuited		-	-	36.4	nF
C_{oes}	Output capacitance			-	-	1.0	
C_{res}	Reverse transfer capacitance			-	-	0.4	
Q_G	Gate charge	V_{CC}=600 V, I_C=150 A, V_{GE}=15 V		-	1.13	-	μC
$t_{d(on)}$	Turn-on delay time	V_{CC}=600 V, I_C=150 A, V_{GE}=±15 V, R_G=0 Ω, Inductive load		-	-	500	ns
t_r	Rise time			-	-	150	
$t_{d(off)}$	Turn-off delay time			-	-	500	
t_f	Fall time			-	-	400	
V_{EC} (Note1) (Terminal)	Emitter-collector voltage	I_E=150 A, G-E short-circuited, Refer to the figure of test circuit (Note5)	T_{vj}=25 °C	-	1.90	2.45	V
			T_{vj}=125 °C	-	2.15	-	
			T_{vj}=150 °C	-	2.20	-	
V_{EC} (Note1) (Chip)		I_E=150 A, G-E short-circuited, (Note5)	T_{vj}=25 °C	-	1.55	2.00	V
			T_{vj}=125 °C	-	1.55	-	
			T_{vj}=150 °C	-	1.55	-	
t_{rr} (Note1)	Reverse recovery time	V_{CC}=600 V, I_E=150 A, V_{GE}=±15 V, R_G=0 Ω, Inductive load		-	-	400	ns
Q_{rr} (Note1)	Reverse recovery charge			-	15.0	-	μC
E_{on}	Turn-on switching energy per pulse	V_{CC}=600 V, I_C=I_E=150 A, V_{GE}=±15 V, R_G=0 Ω, T_{vj}=150 °C, Inductive load		-	14.8	-	mJ
E_{off}	Turn-off switching energy per pulse			-	15.3	-	
E_{rr} (Note1)	Reverse recovery energy per pulse			-	8.6	-	mJ
r_g	Internal gate resistance	Per switch		-	2.0	-	Ω

BRAKE PART IGBT/DIODE

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
I_{CES}	Collector-emitter cut-off current	$V_{CE}=V_{CES}$, G-E short-circuited		-	-	1.0	mA
I_{GES}	Gate-emitter leakage current	$V_{GE}=V_{GES}$, C-E short-circuited		-	-	0.5	μA
$V_{GE(th)}$	Gate-emitter threshold voltage	I_C=10 mA, V_{CE}=10 V		5.4	6.0	6.6	V
V_{CEsat} (Terminal)	Collector-emitter saturation voltage	I_C=100 A, V_{GE}=15 V, Refer to the figure of test circuit (Note5)	T_{vj}=25 °C	-	1.75	2.20	V
			T_{vj}=125 °C	-	2.10	-	
			T_{vj}=150 °C	-	2.20	-	
V_{CEsat} (Chip)		I_C=100 A, V_{GE}=15 V, (Note5)	T_{vj}=25 °C	-	1.55	1.80	V
			T_{vj}=125 °C	-	1.75	-	
			T_{vj}=150 °C	-	1.80	-	
C_{ies}	Input capacitance	V_{CE}=10 V, G-E short-circuited		-	-	24.2	nF
C_{oes}	Output capacitance			-	-	0.7	
C_{res}	Reverse transfer capacitance			-	-	0.3	
Q_G	Gate charge	V_{CC}=600 V, I_C=100 A, V_{GE}=15 V		-	0.75	-	μC
$t_{d(on)}$	Turn-on delay time	V_{CC}=600 V, I_C=100 A, V_{GE}=±15 V, R_G=3.9 Ω, Inductive load		-	-	300	ns
t_r	Rise time			-	-	150	
$t_{d(off)}$	Turn-off delay time			-	-	500	
t_f	Fall time			-	-	400	

ELECTRICAL CHARACTERISTICS (cont.; T_{vj}=25 °C, unless otherwise specified)

BRAKE PART IGBT/DIODE

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
E_{on}	Turn-on switching energy per pulse	V_{CC}=600 V, V_{GE}=±15 V, T_{vj}=150 °C, Inductive load	I_C=100A,R_G=3.9Ω	-	4.1	-	mJ
E_{off}	Turn-off switching energy per pulse			-	10.6	-	
E_{rr}	Reverse recovery energy per pulse		I_E=100A,R_G=3.9Ω	-	7.8	-	
r_g	Internal gate resistance	-		-	0	-	Ω
I_{RRM}	Reverse current	V_R=V_{RRM}, G-E short-circuited		-	-	1.0	mA
V_F (Terminal)	Forward voltage	I_F=75 A, G-E short-circuited, Refer to the figure of test circuit (Note5)	T_{vj}=25 °C	-	1.80	2.40	V
			T_{vj}=125 °C	-	2.00	-	
			T_{vj}=150 °C	-	2.05	-	
V_F (Chip)		I_F=75 A, G-E short-circuited, (Note5)	T_{vj}=25 °C	-	1.65	2.10	V
			T_{vj}=125 °C	-	1.65	-	
			T_{vj}=150 °C	-	1.65	-	
t_{rr}	Reverse recovery time	V_{CC}=600 V, I_F=75 A, V_{GE}=±15 V, R_G=5.6 Ω, Inductive load		-	-	400	ns
Q_{rr}	Reverse recovery charge			-	7.5	-	μC

CONVERTER PART DIODE

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
I_{RRM}	Repetitive peak reverse current	V_R=V_{RRM}, T_{vj}=150 °C		-	-	20	mA
V_F (Terminal)	Forward voltage	I_F=150 A	T_{vj}=25 °C	-	1.40	1.85	V
			T_{vj}=150 °C	-	1.30	-	
V_F (chip)			T_{vj}=25 °C	-	1.15	1.40	
			T_{vj}=150 °C	-	1.10	-	

NTC THERMISTOR PART

Symbol	Item	Conditions	Limits			Unit
			Min.	Typ.	Max.	
R_{25}	Zero-power resistance	T_C=25 °C (Note4)	4.85	5.00	5.15	kΩ
ΔR/R	Deviation of resistance	R_{100}=493 Ω, T_C=100 °C (Note4)	-7.3	-	+7.8	%
$B_{(25/50)}$	B-constant	Approximate by equation (Note6)	-	3375	-	K
P_{25}	Power dissipation	T_C=25 °C (Note4)	-	-	10	mW

THERMAL RESISTANCE CHARACTERISTICS

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
$R_{th(j-c)Q}$	Thermal resistance	Junction to case, per Inverter IGBT (Note4)		-	-	197	K/kW
$R_{th(j-c)D}$		Junction to case, per Inverter FWD (Note4)		-	-	391	
$R_{th(j-c)Q}$		Junction to case, Brake IGBT (Note4)		-	-	264	
$R_{th(j-c)D}$		Junction to case, Brake DIODE (Note4)		-	-	628	
$R_{th(j-c)D}$		Junction to case, per Converter DIODE (Note4)				324	
$R_{th(c-s)}$	Contact thermal resistance	Case to heat sink, per 1 module,	Thermal grease applied (Note4, 7, 10)	-	11.5	-	K/kW
			PC-TIM applied (Note4, 8, 10)	-	3.1	-	

MECHANICAL CHARACTERISTICS

Symbol	Item	Conditions		Limits			Unit
				Min.	Typ.	Max.	
M_s	Mounting torque	Mounting to heat sink	M 5 screw	2.5	3.0	3.5	N·m
d_s	Creepage distance	Solder pin type(MXUD)	Terminal to terminal	11.7	-	-	mm
			Terminal to base plate	18.3	-	-	
		Pressfit pin type(MXUDP)	Terminal to terminal	5.1	-	-	
			Terminal to base plate	15.8	-	-	
d_a	Clearance	Solder pin type(MXUD)	Terminal to terminal	6.5	-	-	
			Terminal to base plate	18.1	-	-	
		Pressfit pin type(MXUDP)	Terminal to terminal	5.0	-	-	
			Terminal to base plate	15.8	-	-	
e_c	Flatness of base plate	On the centerline X, Y (Note9)		±0	-	+200	μm
m	mass	-		-	270	-	g

RECOMMENDED OPERATING CONDITIONS

Symbol	Item	Conditions	Limits			Unit
			Min.	Typ.	Max.	
V_{CC}	(DC) Supply voltage	Applied across P-N(P1-N1) terminals	-	600	850	V
V_{GEon}	Gate (-emitter drive) voltage	Applied across G*P-*/G*N-E/GB-E terminals (*=U,V,W)	13.5	15.0	16.5	V
R_G	External gate resistance	Inverter IGBT, Per switch	0	-	20	Ω
		Brake IGBT	3.9	-	40	

*: This product is compliant with the Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment (RoHS) directive 2011/65/EU and (EU) 2015/863.

Note1. Represent ratings and characteristics of the anti-parallel, emitter-collector free-wheeling diode (FWD).

2. Junction temperature (T_{vj}) should not increase beyond T_{vjmax} rating.

3. Pulse width and repetition rate should be such that the device junction temperature (T_{vj}) dose not exceed T_{vjmax} rating.

4. Case temperature (T_C) and heat sink temperature (T_S) are defined on the each surface (mounting side) of base plate and heat sink just under the chips. Refer to the figure of chip location.

5. Pulse width and repetition rate should be such as to cause negligible temperature rise. Refer to the figure of test circuit.

6. $B_{(25/50)} = \ln(\frac{R_{25}}{R_{50}})/(\frac{1}{T_{25}} - \frac{1}{T_{50}})$

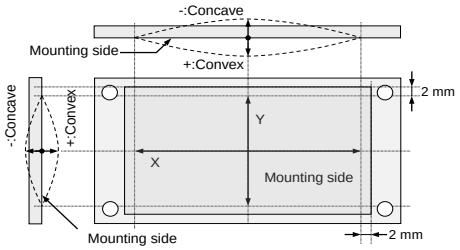
   R_{25}: resistance at absolute temperature T_{25} [K]; T_{25}=25 [°C]+273.15=298.15 [K]

   R_{50}: resistance at absolute temperature T_{50} [K]; T_{50}=50 [°C]+273.15=323.15 [K]

7. Typical value is measured by using thermally conductive grease of λ=0.9 W/(m·K)/$D_{(C-S)}$=50 μm.

8. Typical value is measured by using PC-TIM of λ=3.4 W/(m·K)/$D_{(C-S)}$=50 μm.

9. The base plate (mounting side) flatness measurement points (X, Y) are shown in the following figure.



10. Long term performance related to thermal conductive grease and PC-TIM (including but not limited to aspects such as the increase of thermal resistance due to pumping out, etc.) should be verified under your specific application conditions. Each temperature condition ($T_{vj\,max}$, $T_{vj\,op}$, $T_{C\,max}$) must be maintained below the maximum rated temperature throughout consideration of the temperature rise even for long term usage.

11. Use the following screws when mounting the printed circuit board (PCB) on the standoffs.

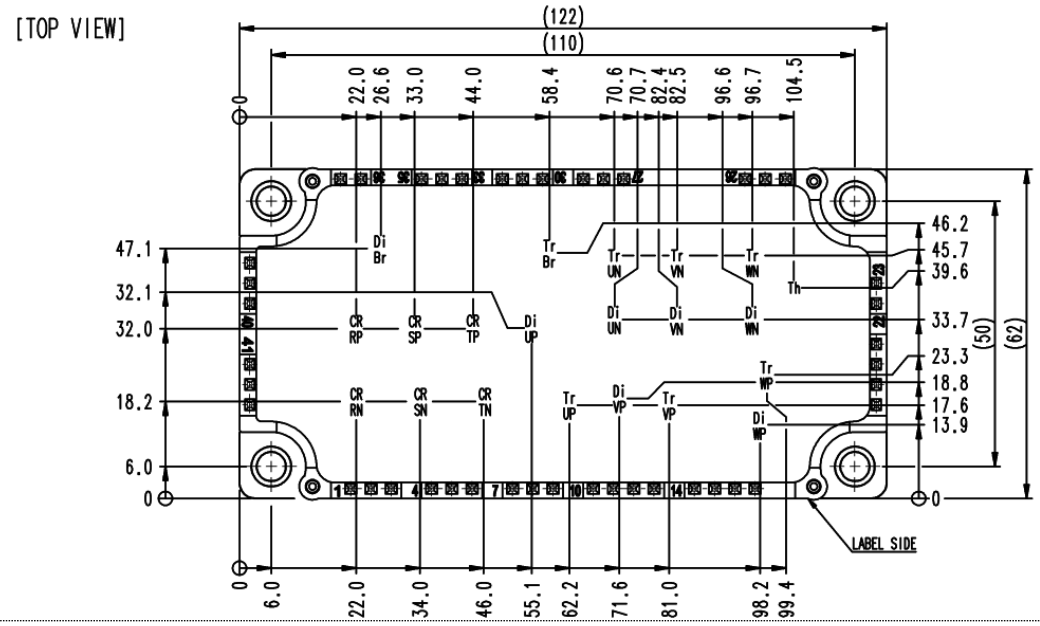
PCB thickness : t1.6

	Type	Manufacturer	Size	Tightening torque (N・m)	Recommended tightening method
(1)	PT®	EJOT	K25×8	0.55 ± 0.055	by handwork (equivalent to 30 rpm by mechanical screw driver) ~ 600 rpm (by mechanical screw driver)
(2)	PT®		K25×10	0.75 ± 0.075 N・m	
(3)	DELTA PT®		25×8	0.55 ± 0.055 N・m	
(4)	DELTA PT®		25×10	0.75 ± 0.075 N・m	
(5)	B1 tapping screw	-	φ2.6×10	0.75 ± 0.075 N・m	
			φ2.6×12		

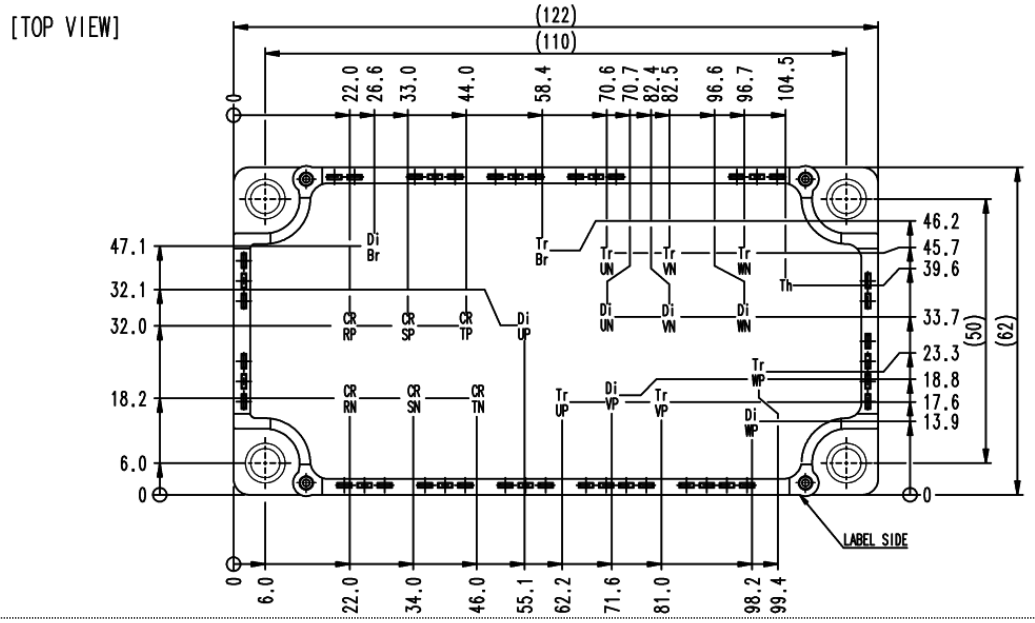
CHIP LOCATION (Top view)

Dimension in mm, tolerance: ±1 mm

MXUD

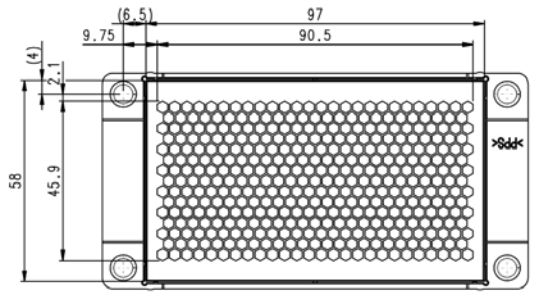
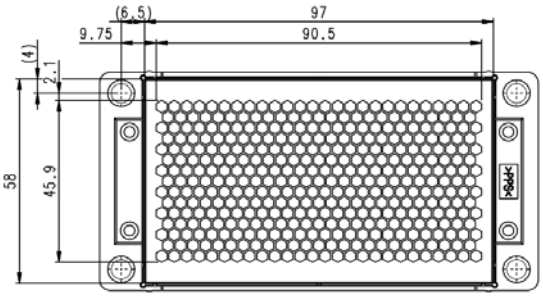


MXUDP

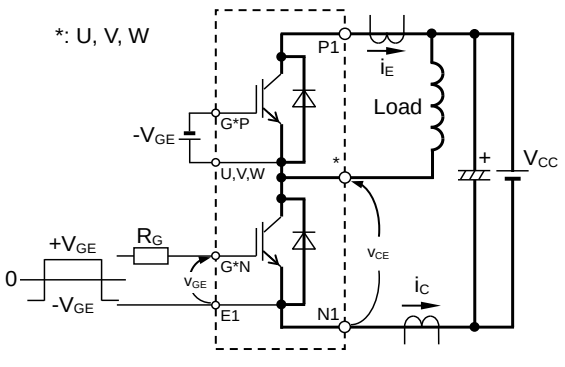


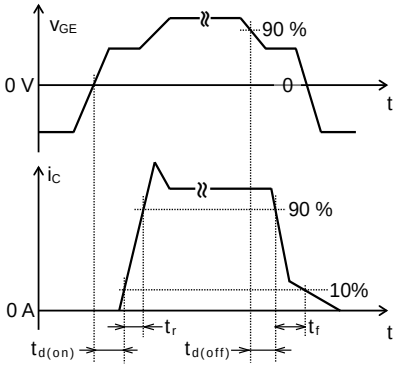
Tr*P/Tr*N/TrBr: IGBT, Di*P/Di*N: DIODE (*=U/V/W), DiBr: BRAKE DIODE,
CR*P/CR*N: CONVERTER DIODE (*=R/S/T), Th: NTC thermistor

Option: PC-TIM applied baseplate outline

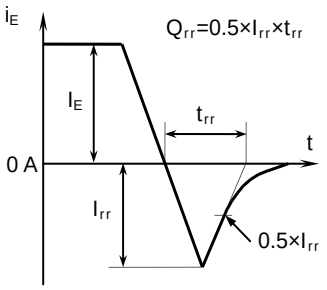
MXUD	MXUDP
	

TEST CIRCUIT AND WAVEFORMS

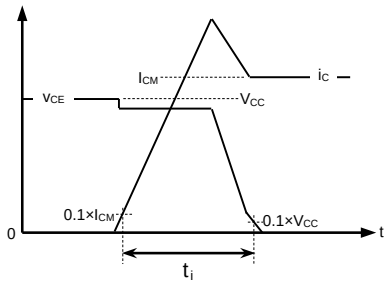




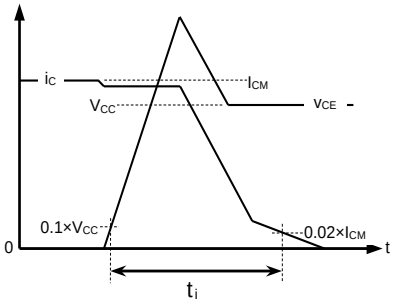
Switching characteristics test circuit and waveforms



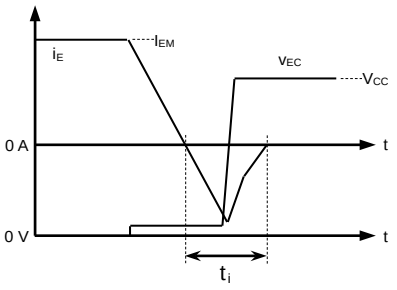
t_{rr}, Q_{rr} characteristics test waveform



IGBT Turn-on switching energy



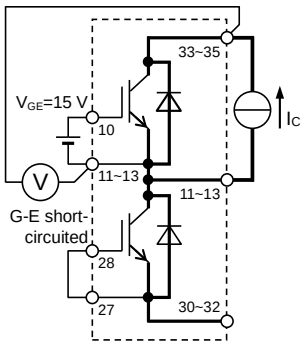
IGBT Turn-off switching energy



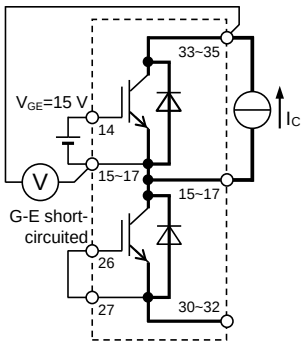
FWD Reverse recovery energy

Switching energy and Reverse recovery energy test waveforms (Integral time instruction drawing)

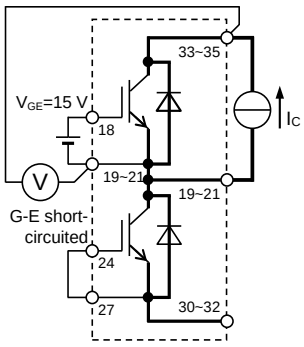
TEST CIRCUIT



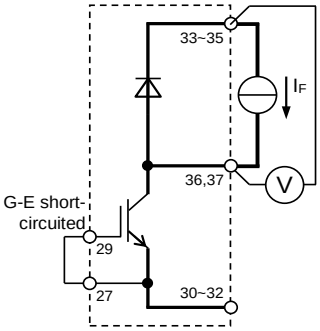
TrUP



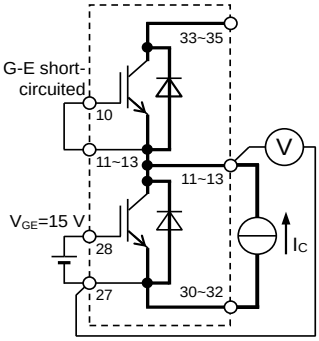
TrVP



TrWP

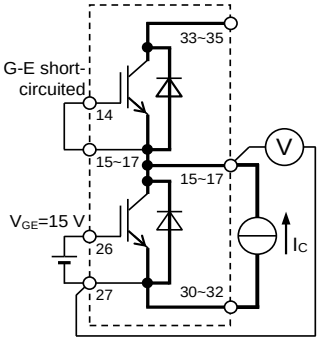


Brake DIODE



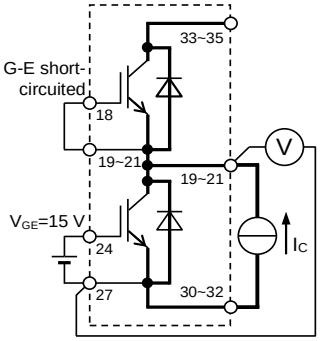
TrUN

Gate-emitter short-circuited
GVP-V, GVN-E1, GWP-W, GWN-E1 GB-E1



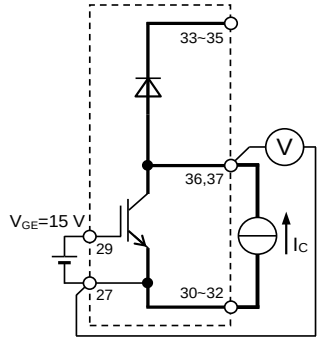
TrVN

Gate-emitter short-circuited
GUP-U, GUN-E1, GWP-W, GWN-E1 GB-E1



TrWN

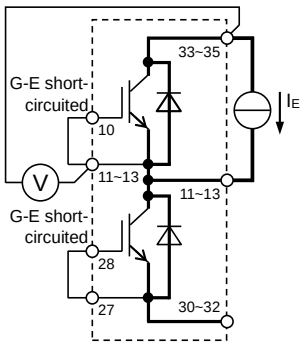
Gate-emitter short-circuited
GUP-U, GUN-E1, GVP-V, GVN-E1 GB-E1



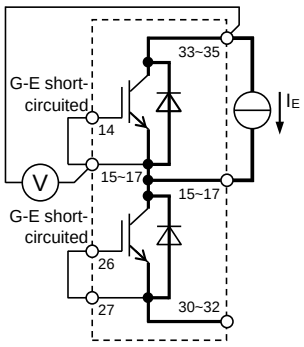
Brake IGBT

Gate-emitter short-circuited
GUP-U, GUN-E1, GVP-V, GVN-E1, GWP-W, GWN-E1

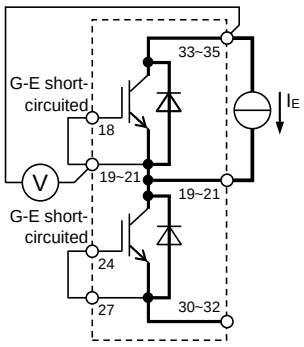
V_{CEsat} /BRAKE DIODE V_F characteristics test circuit



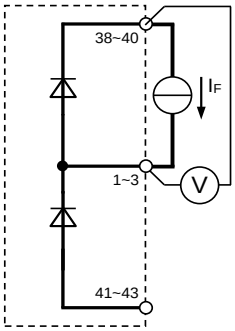
DiUP

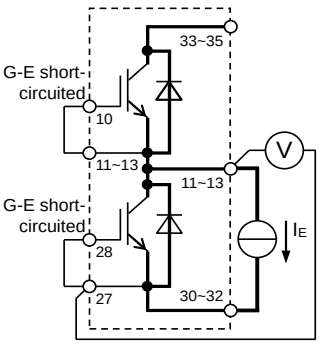


DiVP



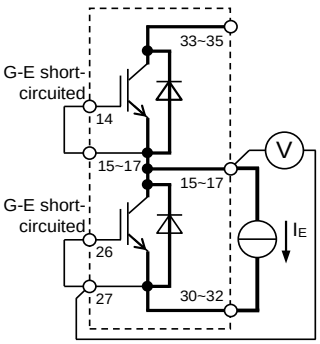
DiWP





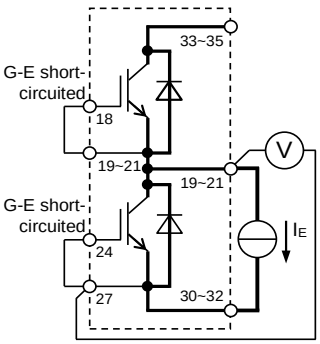
DiUN

Gate-emitter short-circuited
GVP-V, GVN-E1, GWP-W, GWN-E1 GB-E1



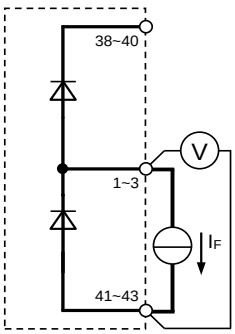
DiVN

Gate-emitter short-circuited
GUP-U, GUN-E1, GWP-W, GWN-E1 GB-E1



DiWN

Gate-emitter short-circuited
GUP-U, GUN-E1, GVP-V, GVN-E1 GB-E1



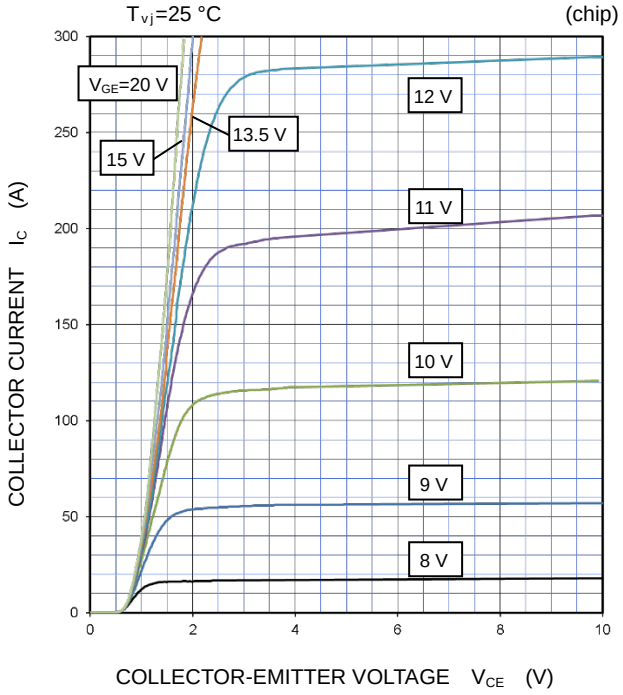
CONVERTER DIODE (ex.phase-R)

V_{EC} / CONVERTER DIODE V_F characteristics test circuit

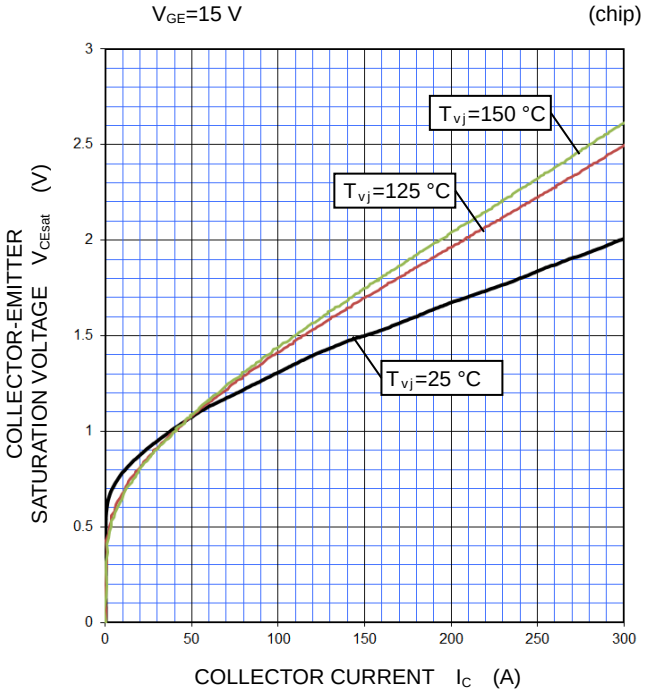
PERFORMANCE CURVES

INVERTER PART

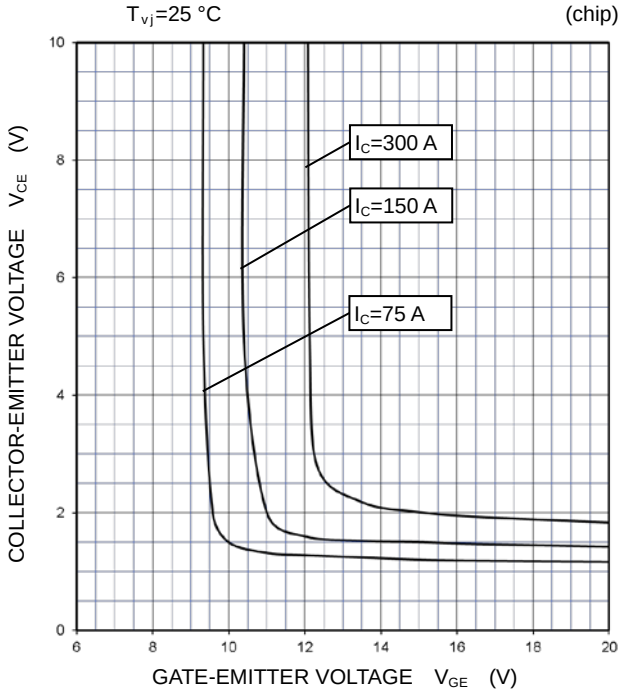
OUTPUT CHARACTERISTICS (TYPICAL)



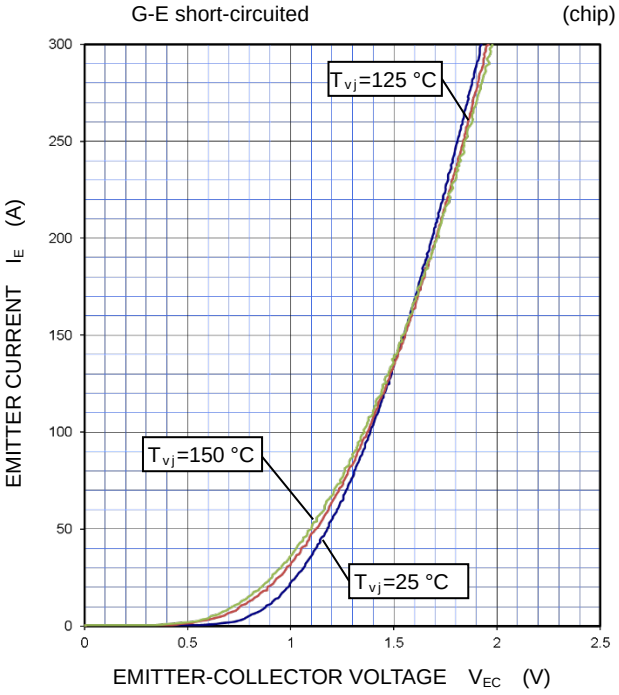
COLLECTOR-EMITTER SATURATION VOLTAGE CHARACTERISTICS (TYPICAL)



COLLECTOR-EMITTER VOLTAGE CHARACTERISTICS (TYPICAL)



FREE WHEELING DIODE FORWARD CHARACTERISTICS (TYPICAL)

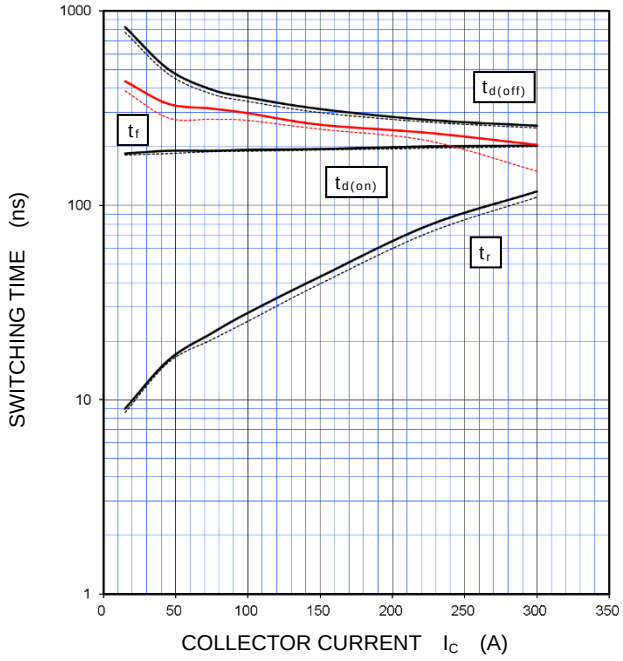


PERFORMANCE CURVES

INVERTER PART

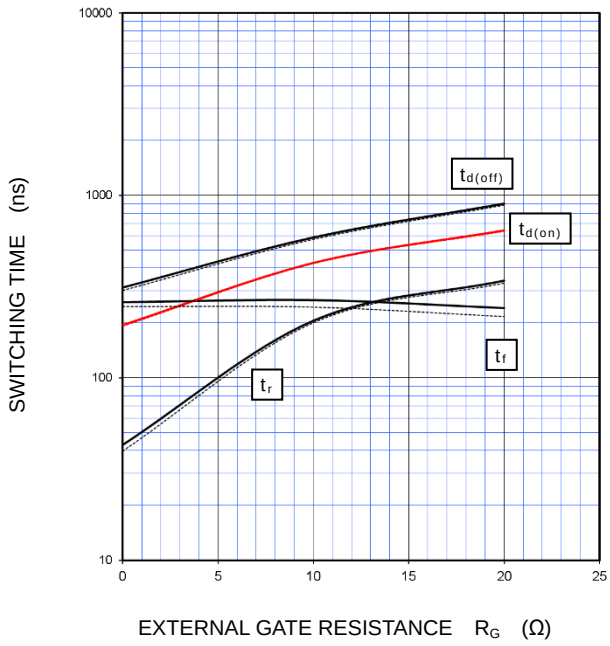
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, R_G=0 Ω, V_{GE}=±15 V, INDUCTIVE LOAD
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C



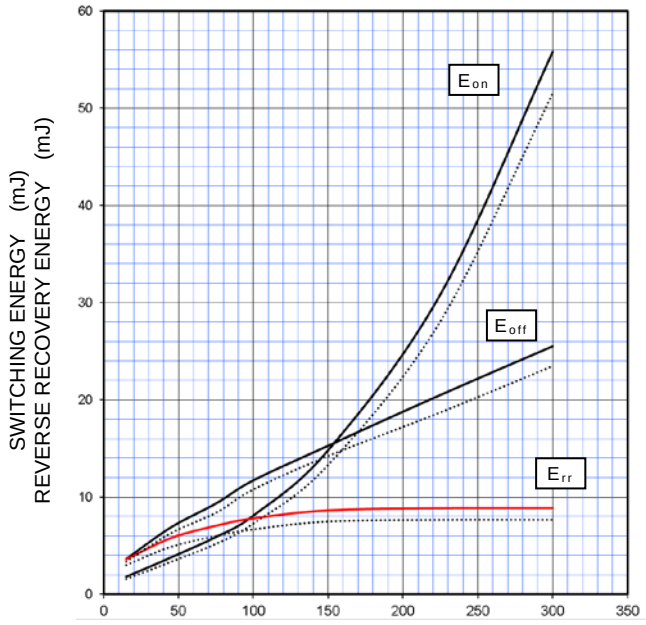
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C=150 A, V_{GE}=±15 V, INDUCTIVE LOAD
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C



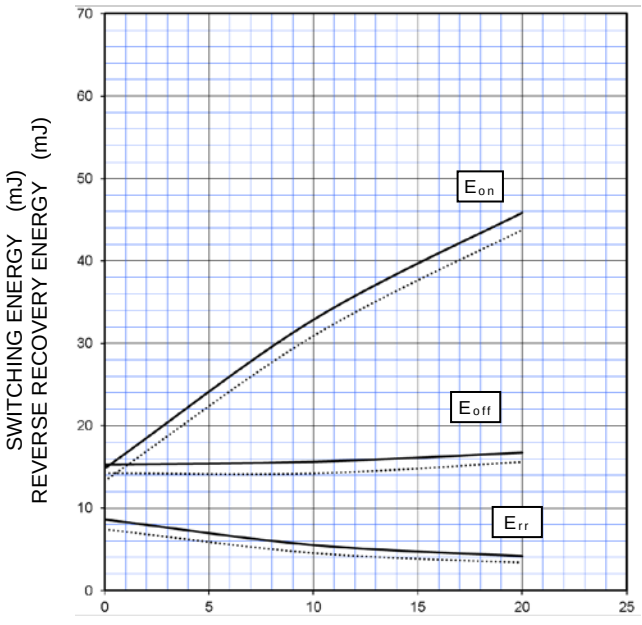
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, R_G=0 Ω, V_{GE}=±15 V, INDUCTIVE LOAD,
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C, PER PULSE



HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C/I_E=150 A, V_{GE}=±15 V, INDUCTIVE LOAD,
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C, PER PULSE

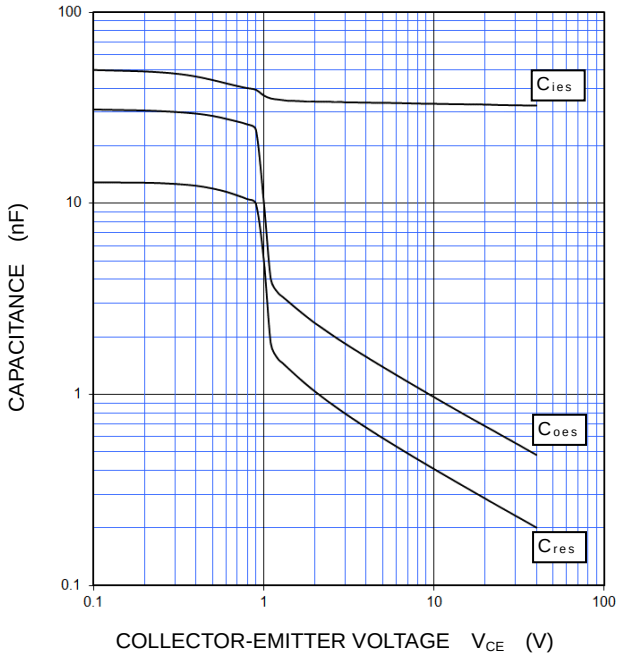


PERFORMANCE CURVES

INVERTER PART

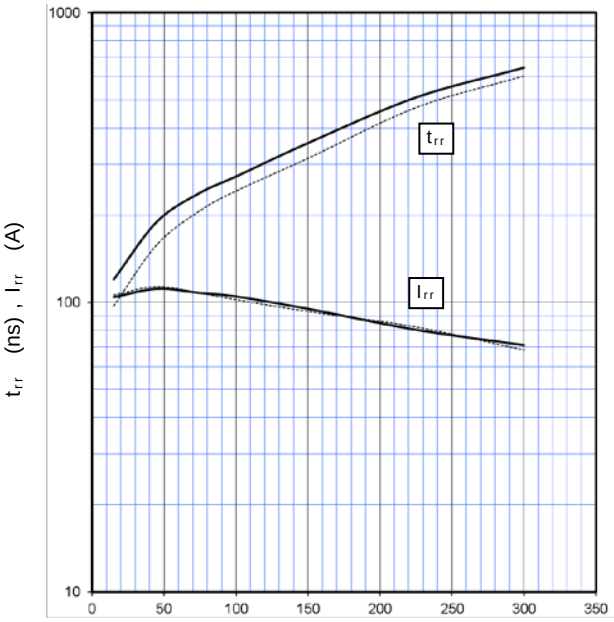
CAPACITANCE CHARACTERISTICS (TYPICAL)

G-E short-circuited, T_{vj}=25 °C



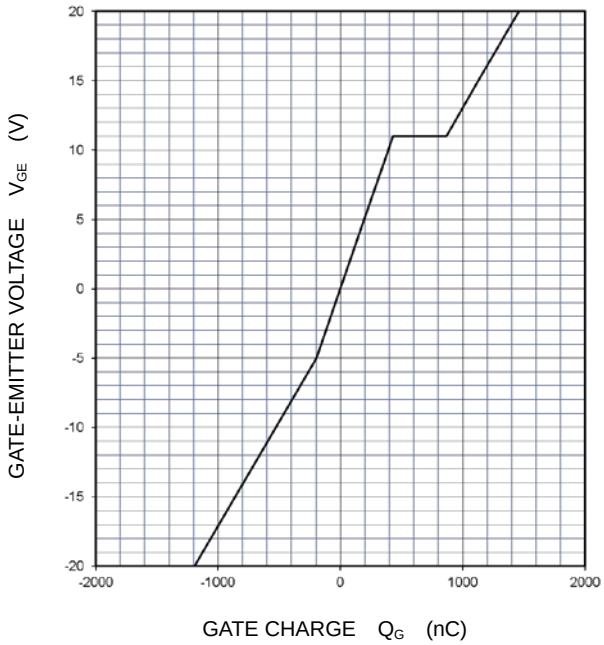
FREE WHEELING DIODE REVERSE RECOVERY CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, R_G=0 Ω, V_{GE}=±15 V, INDUCTIVE LOAD
———: T_j=150 °C, - - - - -: T_j=125 °C



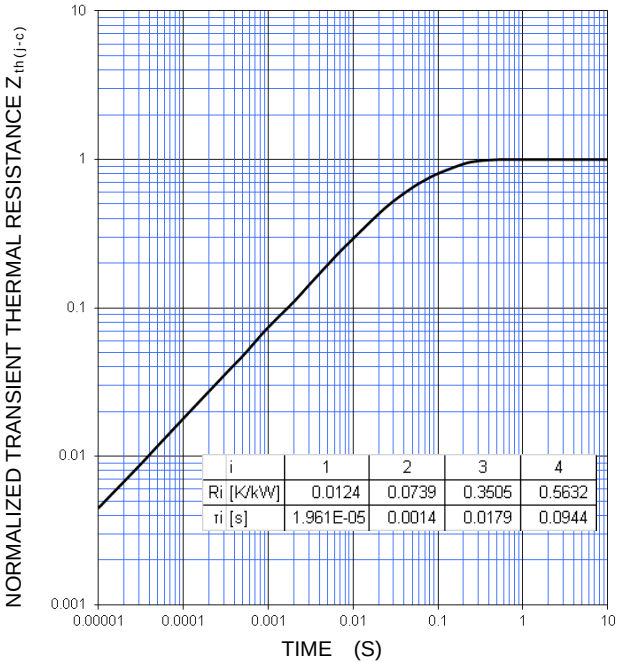
GATE CHARGE CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C=150 A, T_{vj}=25 °C



TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS (MAXIMUM)

Single pulse, T_C=25 °C
$R_{th(j-c)Q}$=197 K/kW, $R_{th(j-c)D}$=391 K/kW



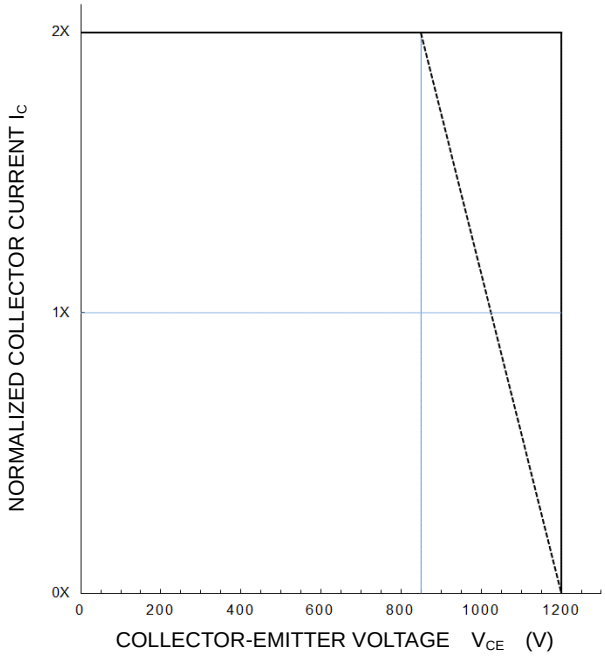
i	1	2	3	4
Ri [K/kW]	0.0124	0.0739	0.3505	0.5632
ri [s]	1.961E-05	0.0014	0.0179	0.0944

PERFORMANCE CURVES

INVERTER PART

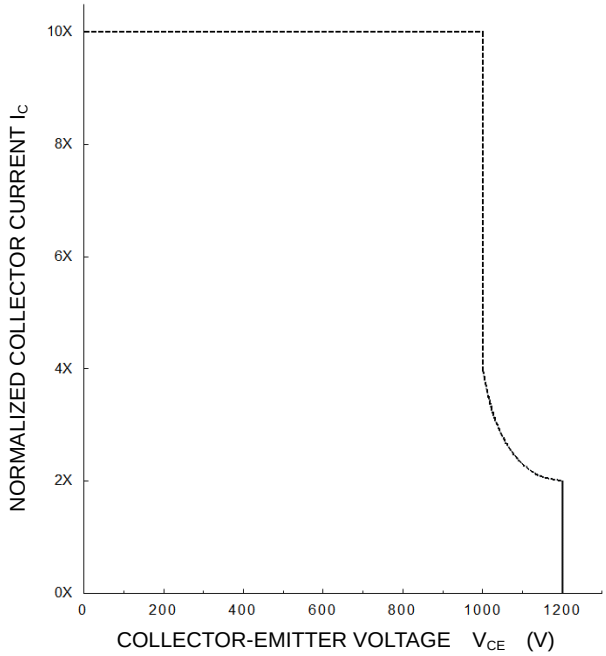
TURN-OFF SWITCHING SAFE OPERATIONG AREA (REVERSE BIAS SAFE OPERATING AREA) (MAXIMUM)

V_{CC}≤850 V, R_G=0~20 Ω, V_{GE}=±15 V,
————: T_{vj}=25~150 °C (Normal load operations (Continuous)
- - - - - -: T_{vj}=175 °C (Unusual load operations (Limited period)



SHORT-CIRCUIT SAFE OPERATING AREA (MAXIMUM)

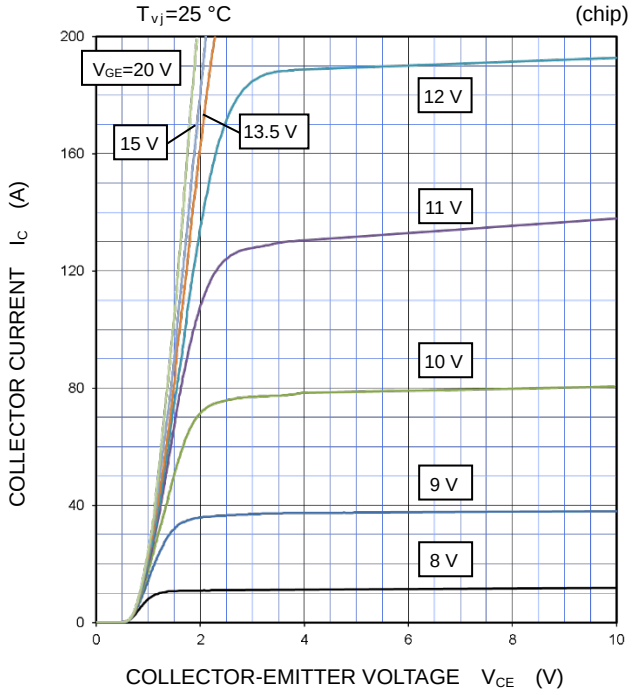
V_{CC}≤800 V, R_G=0~20 Ω, V_{GE}=±15 V,
T_{vj}= 25 ~ 150 °C, t_W≤8 μs, Non-Repetitive



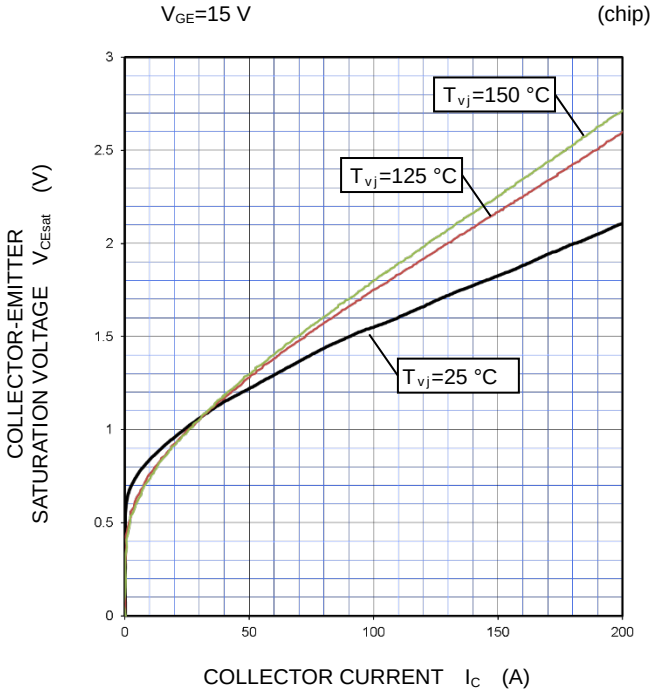
PERFORMANCE CURVES

BRAKE PART

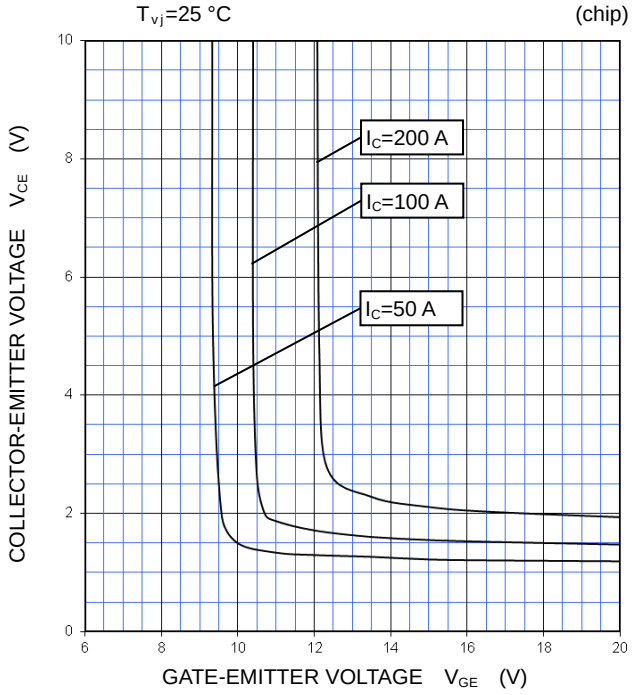
OUTPUT CHARACTERISTICS (TYPICAL)



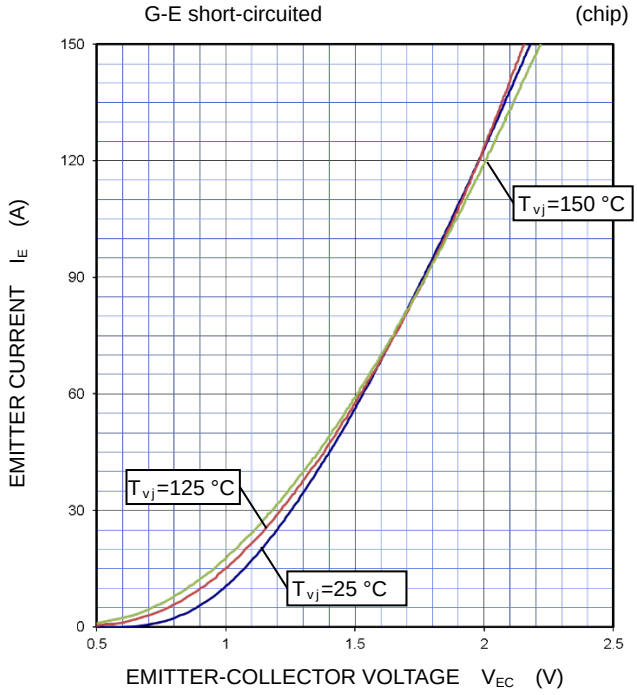
COLLECTOR-EMITTER SATURATION VOLTAGE CHARACTERISTICS (TYPICAL)



COLLECTOR-EMITTER VOLTAGE CHARACTERISTICS (TYPICAL)



DIODE FORWARD CHARACTERISTICS (TYPICAL)

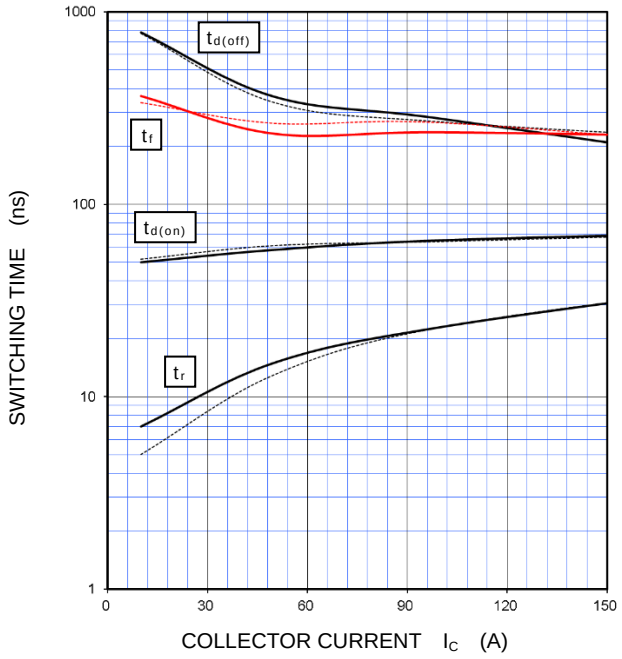


PERFORMANCE CURVES

BRAKE PART

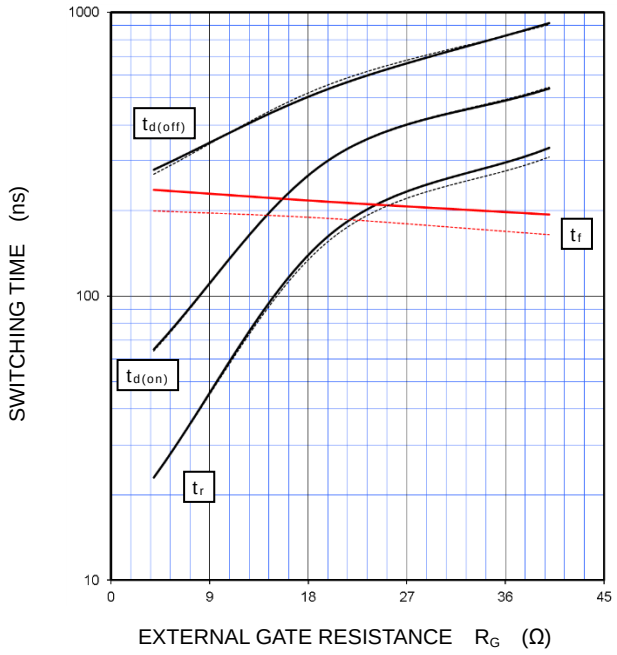
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, R_G=3.9 Ω, V_{GE}=±15 V, INDUCTIVE LOAD
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C



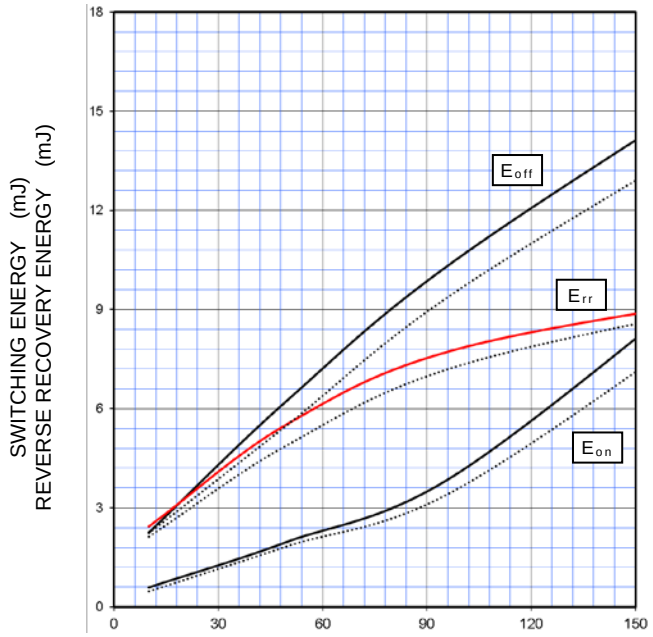
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C=100 A, V_{GE}=±15 V, INDUCTIVE LOAD
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C



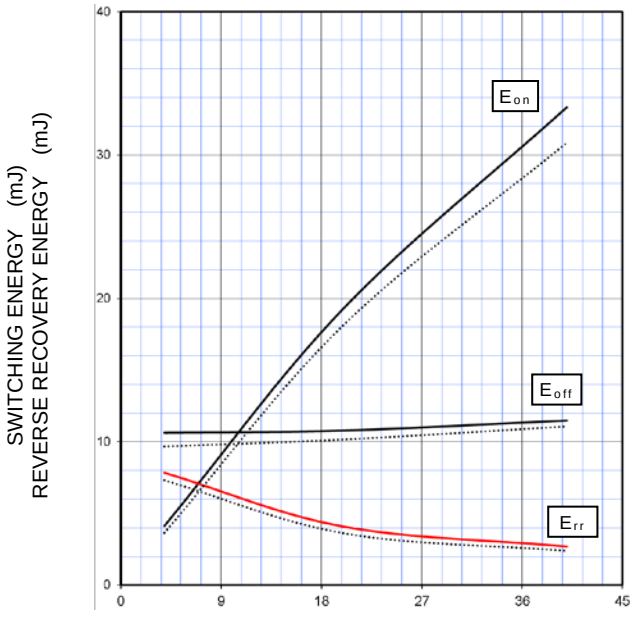
HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, R_G=3.9 Ω, V_{GE}=±15 V, INDUCTIVE LOAD,
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C, PER PULSE



HALF-BRIDGE SWITCHING CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C/I_E=100 A, V_{GE}=±15 V, INDUCTIVE LOAD,
————: T_{vj}=150 °C, - - - - -: T_{vj}=125 °C, PER PULSE

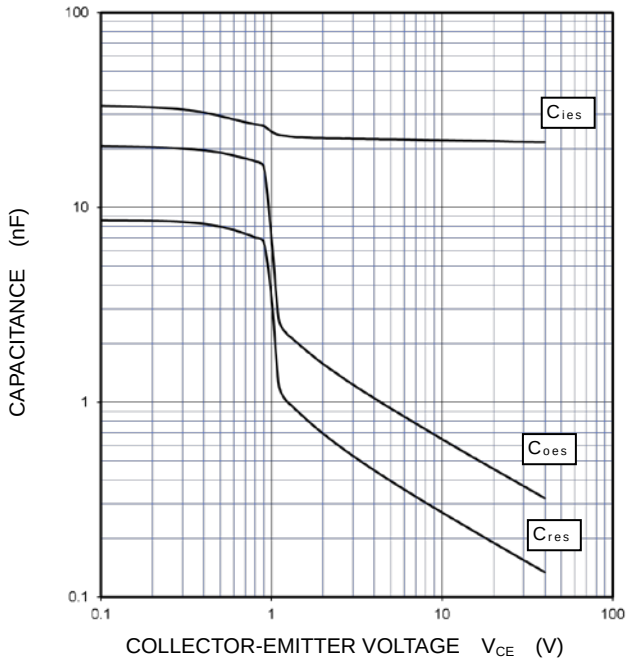


PERFORMANCE CURVES

BRAKE PART

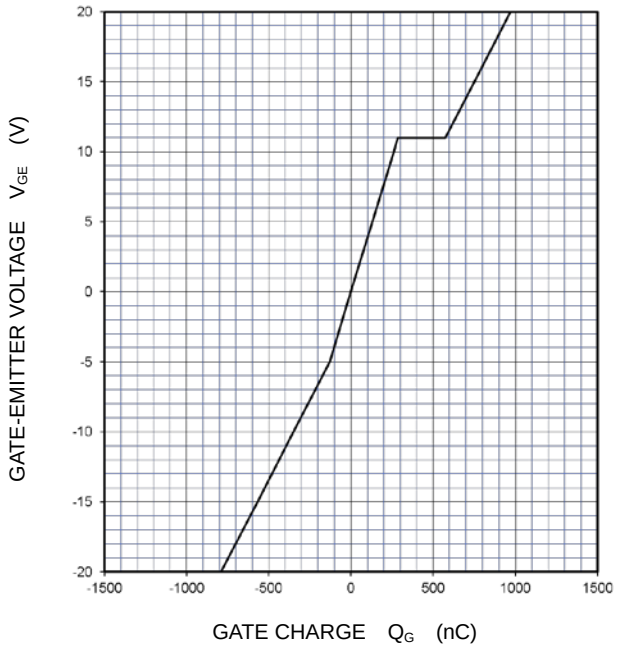
CAPACITANCE CHARACTERISTICS (TYPICAL)

G-E short-circuited, T_{vj}=25 °C



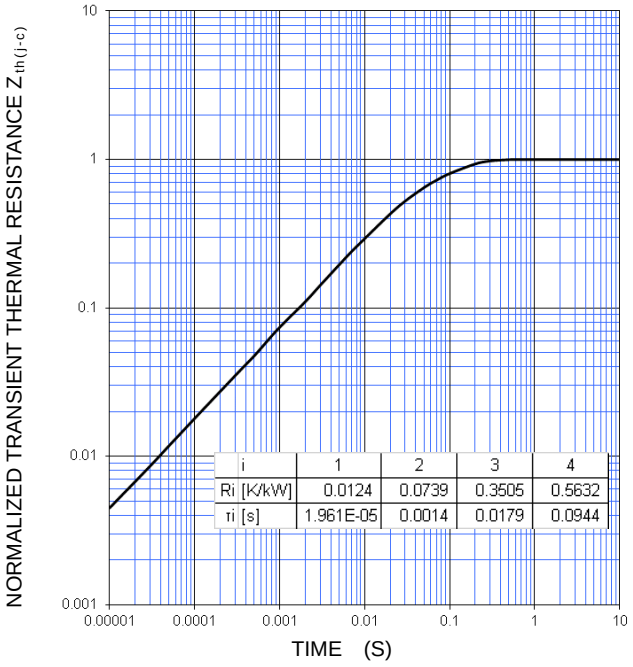
GATE CHARGE CHARACTERISTICS (TYPICAL)

V_{CC}=600 V, I_C=100 A, T_{vj}=25 °C



TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS (MAXIMUM)

Single pulse, T_C=25 °C
$R_{th(j-c)Q}$=264 K/kW, $R_{th(j-c)D}$=628 K/kW



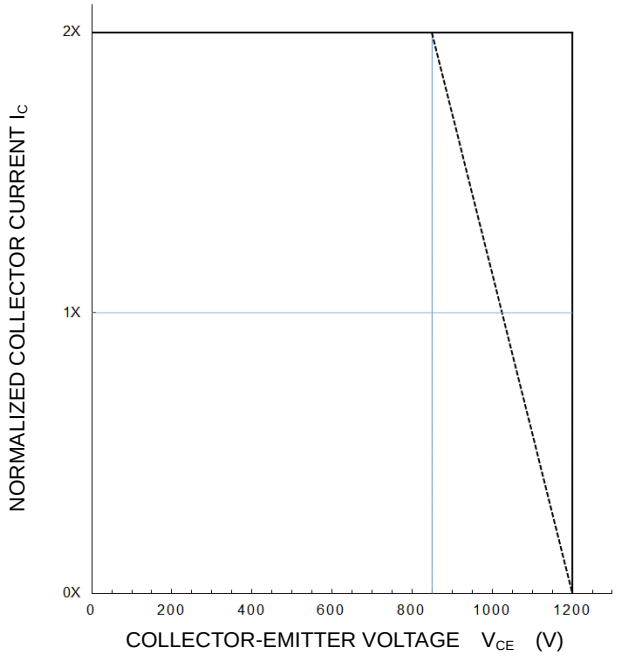
i	1	2	3	4
Ri [K/kW]	0.0124	0.0739	0.3505	0.5632
τi [s]	1.961E-05	0.0014	0.0179	0.0944

PERFORMANCE CURVES

BRAKE PART

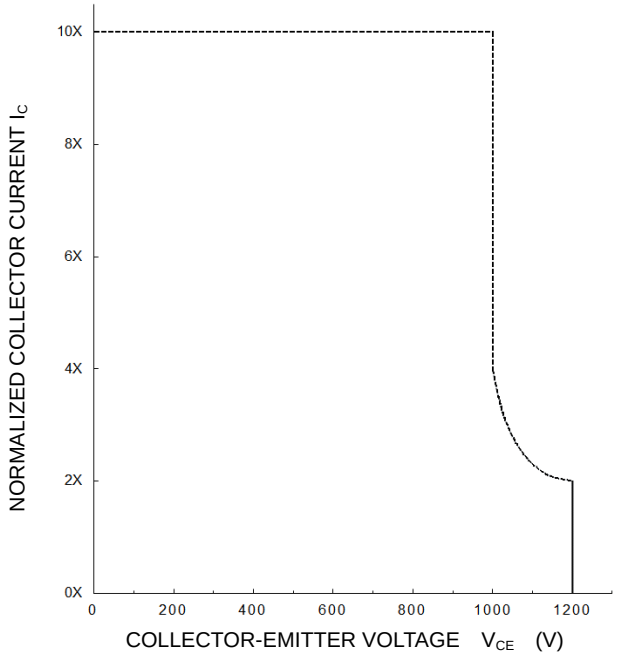
TURN-OFF SWITCHING SAFE OPERATIONG AREA (REVERSE BIAS SAFE OPERATING AREA) (MAXIMUM)

V_{CC}≤850 V, R_G=3.9~40 Ω, V_{GE}=±15 V,
————: T_{vj}=25~150 °C (Normal load operations (Continuous)
- - - - - -: T_{vj}=175 °C (Unusual load operations (Limited period)



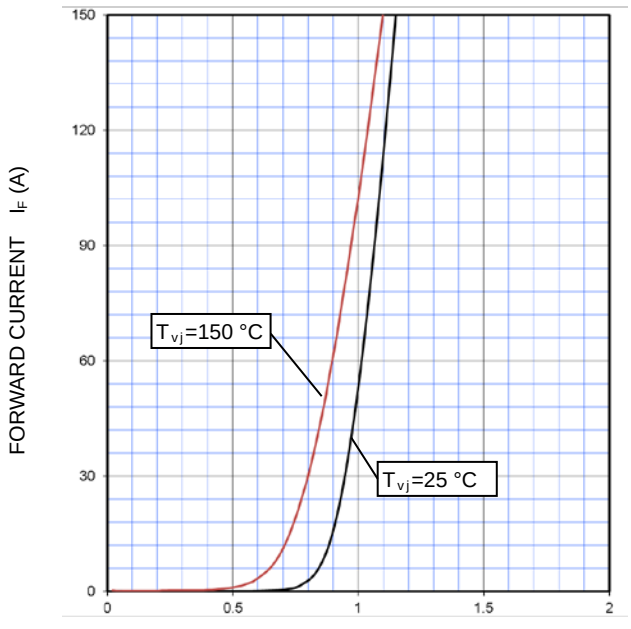
SHORT-CIRCUIT SAFE OPERATING AREA (MAXIMUM)

V_{CC}≤800 V, R_G=3.9~40 Ω, V_{GE}=±15 V,
T_{vj}= 25 ~ 150 °C, t_W≤8 μs, Non-Repetitive



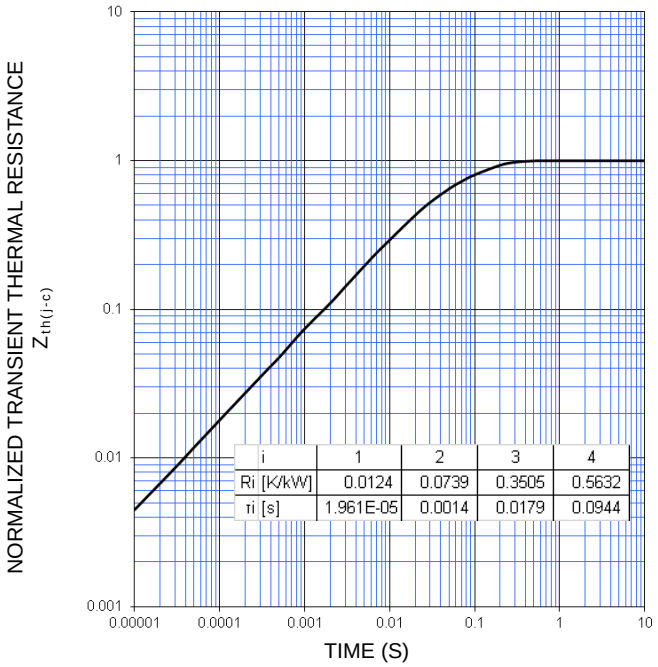
CONVERTER PART

CONVERTER DIODE FORWARD CHARACTERISTICS (TYPICAL)



TRANSIENT THERMAL IMPEDANCE CHARACTERISTICS (MAXIMUM)

Single pulse, T_C=25 °C
$R_{th(j-c)D}$=324 K/kW

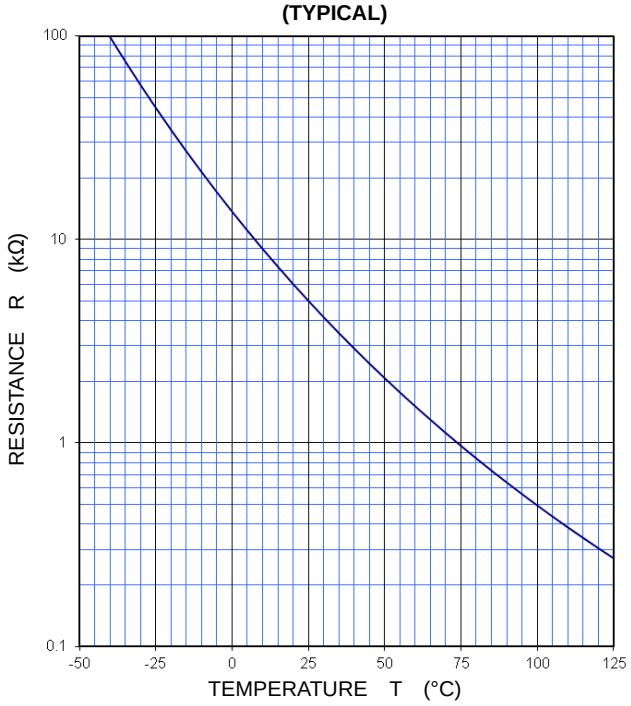


i	1	2	3	4
Ri [K/kW]	0.0124	0.0739	0.3505	0.5632
τi [s]	1.961E-05	0.0014	0.0179	0.0944

PERFORMANCE CURVES

NTC thermistor part

TEMPERATURE CHARACTERISTICS (TYPICAL)



Note: The characteristics curves are presented for reference only and not guaranteed by production test, unless otherwise noted.

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